

Jinwon Kim

Robotics Engineer

📍 Seoul, South Korea
 ✉️ mqjinwon@gmail.com
 ☎️ 010-5068-7329
 🌐 mqjinwon.github.io
 📄 robotics-jinwon
 📱 mqjinwon

Summary

- 3+ years of robotics engineering experience with proven track record in national R&D projects (ADD, NIPA, IITP, COMPA, KOITA, GERI), delivering production-grade solutions from concept to deployment.
- Full-stack robotics engineer specializing in perception, control, and system integration with hands-on experience building and deploying real-world robotic systems that bridge hardware and software.
- Engineering-driven problem solver who delivers reliable, production-oriented code aligned with business goals, comfortable owning complex technical challenges end-to-end.

Experience

- Korea Robot Manufacturing** [🔗](#), Robotics Engineer (Robot-Intelligence Team Lead) May 2023 – present
2 years 8 months
- Navigation for Quadruped robot
 - Simultaneous Localization and Mapping (SLAM)
- Korea Institute of Science and Technology** [🔗](#), Student Researcher May 2019 – Dec 2019
8 months
- Designed and implemented a data collection pipeline using crawling to acquire 1,000 annotated images of objects in various environments
 - Developed object detection and tracking algorithms using YOLOv3 and Siamese network
 - Published KRoC paper and Registered patent

Education

- PhD** **Korea University**, Computer Science Seoul, South Korea
Sept 2025 – present
- Research area: VLM, RL, Quadruped robot
- MS** **Korea Advanced Institute of Science and Technology**, Robotics Daejeon, South Korea
Mar 2021 – Feb 2023
- Master's in Robotics Program at Scalable Graphics, Vision, and Robotics Lab
- Track: Reinforcement Learning, Deep Learning, Intelligent Robotics
 - GPA: 3.65 / 4.3
- BS** **Kwangwoon University**, Division of Robotics Seoul, South Korea
Mar 2017 – Feb 2021
- Track: Robot Control, Robot Navigation, Computer Vision
 - GPA: 4.23 / 4.5

Publications

- Collision-Backpropagation based Obstacle Avoidance Method for a Legged Robot Expressed as a Simplified Dynamics Model** Jan 2022
Jinwon Kim, Heechan Shin, S. Y.
 (International Conference on Control, Automation and Systems (ICCAS 2022), BEXCO, Busan, Korea)
- Collision Backpropagation-based Obstacle Avoidance Method for a Legged Robot with Simplified Dynamics Model** Jan 2022
Jinwon Kim, Heechan Shin, S. Y.
 (Korea Robotics Society Annual Conference (KRoC 2022), Pyeongchang, Korea)
- Robust Multi Object Detection Using Siamese Network** Jan 2020
Jinwon Kim, KangGeon Kim
 (Korea Robotics Society Annual Conference (KRoC 2020), Pyeongchang, Korea)

Patents

Robust Multi-object Detection Apparatus and Method Using Siamese Network

2020

KangGeon Kim, Jinwon Kim

Projects

Development of Quadrupedal Robot System Technology for Monitoring, Reconnaissance, and Search Missions

National defense research project

Agency for Defense Development (ADD)
Mar 2021 – Jan 2023

- Generated the initial trajectory for trajectory optimization using a deep learning network, resulting in a speedup of up to 100 times

Activities

Auturbo

Mar 2023 – present

Quadruped Robot Team leader (Mar 2024 – ongoing), Regular Member (Mar 2023 – Dec 2023)

- Developing StrideSim
 - A Quadruped Robot Simulation using IsaacSim

BARAM, Kwangwoon University

Seoul, South Korea

Vice President (Jan 2019 – Dec 2019), Regular Member (Mar 2018 – Dec 2018, Jan 2020 – Feb 2021)

Mar 2018 – Feb 2021

- Represented over 60 active members as an elected by members
- Created and showcased six robotic pieces

Honors

- Military Robotics Society Excellence Award (Dec 2025)
- Open SW mini hackathon 3rd Prize (Nov 2020)
- Dean's list (Dec 2020, Jun 2019, Dec 2018)

Skills

Programming Languages: C++, Python, LaTeX

Frameworks & Tools: ROS, PyTorch, CasADi

Language Fluency: Intermediate high in English (TOEIC 810, OPIc IH), Native in Korean